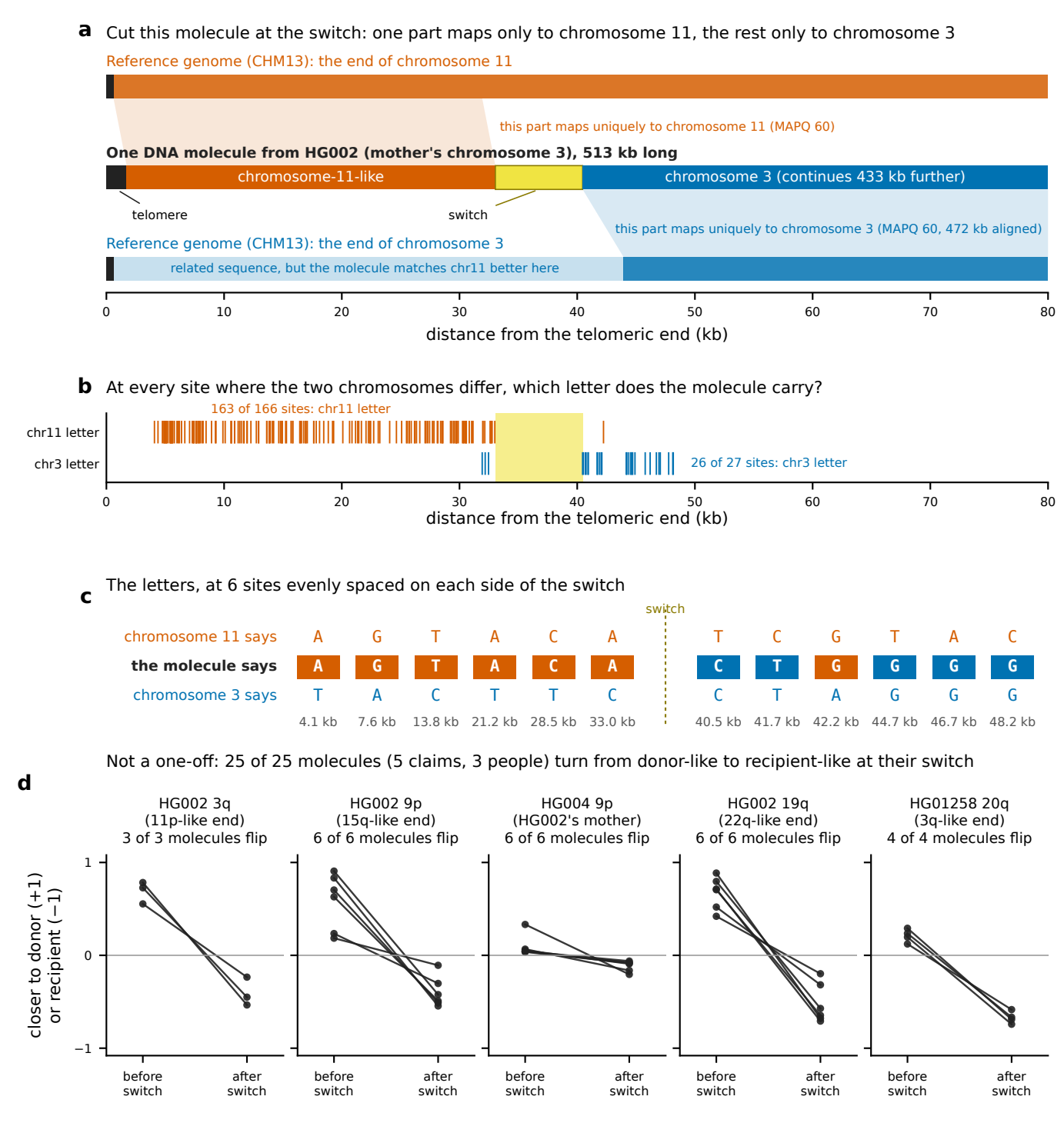
a
Cut this molecule at the switch: one part maps only to chromosome 11, the rest only to chromosome 3
Reference genome (CHM13): the end of chromosome 11
this part maps uniquely to chromosome 11 (MAPQ 60)
One DNA molecule from HG002 (mother's chromosome 3), 513 kb long
chromosome-11-like
chromosome 3 (continues 433 kb further)
telomere
switch
this part maps uniquely to chromosome 3 (MAPQ 60, 472 kb aligned)
Reference genome (CHM13): the end of chromosome 3
related sequence, but the molecule matches chr11 better here
0
10
20
30
40
50
60
70
80
distance from the telomeric end (kb)
b
At every site where the two chromosomes differ, which letter does the molecule carry?
163 of 166 sites: chr11 letter
chr11 letter
chr3 letter
26 of 27 sites: chr3 letter
0
10
20
30
40
50
60
70
80
distance from the telomeric end (kb)
c
The letters, at 6 sites evenly spaced on each side of the switch
switch
chromosome 11 says
the molecule says
chromosome 3 says
A G T A C A
A G T A C A
T A C T T C
T C G T A C
C T G G G G
C T A G G G
4.1 kb
7.6 kb
13.8 kb
21.2 kb
28.5 kb
33.0 kb
40.5 kb
41.7 kb
42.2 kb
44.7 kb
46.7 kb
48.2 kb
d
Not a one-off: 25 of 25 molecules (5 claims, 3 people) turn from donor-like to recipient-like at their switch
HG002 3q
(11p-like end)
3 of 3 molecules flip
HG002 9p
(15q-like end)
6 of 6 molecules flip
HG004 9p
(HG002's mother)
6 of 6 molecules flip
HG002 19q
(22q-like end)
6 of 6 molecules flip
HG01258 20q
(3q-like end)
4 of 4 molecules flip
closer to donor (+1)
or recipient (−1)
1
0
−1
before switch
after switch